What is the aim of EcoTubeHQ?

YouTube users view a whopping 1 billion hours of video each day. That accounts for around 5 billion YouTube videos being watched each day. [comparitech]

1 billion hours of video streamed each day at 1080p uses a *lot* more data than video streamed at 720p & 480p. The table below shows the details [the data figures are approximate due to the multiple codecs used by YouTube].

One exabyte (EB) is equal to one billion gigabytes (GB)

Resolution	Data per Hour	Total Data (1B hours)	% of 1080p
480p	0.6 GB	0.6 EB	18%
720p	2.0 GB	2.0 EB	61%
1080p	3.3 GB	3.3 EB	100%

[Info derived from eskimo.travel]

480p video would reduce the data transmitted daily by approx 2.7 EB when compared to 1080p, and 720p would reduce it by approx 1.3 EB.

EcoTubeHQ's aim is to reduce data consumption, so it does not offer 1080p video as an output resolution, only 480p and 720p are available. To make up for the reduced resolution the player utilizes AMD FSR [FidelityFX Super Resolution] to produce high quality video output – 720p upscaled with FSR offers the clarity of 1080p.

Figure 1: Comparison of 1080p in a browser [left] with EcoTube 720p [right] – click for full size

AMD FSR works with most GPUs and iGPUs.

Minimal Buffering

In addition to AMD FSR, EcoTubeHQ uses a minimal 10 second buffer during playback to further reduce data consumption. In comparison, YouTube can buffer up to 90 seconds of data. I'll use these numbers to explain how EcotubeHQ minimizes buffering, thus reducing wasted data.

Let's say EcoTubeHQ outputs 31 seconds of a video before playback is stopped. This means three 10 second buffers have been downloaded and output, and the fourth was fully downloaded and stopped after one second, thus leaving nine seconds of wasted A/V data in the buffer.

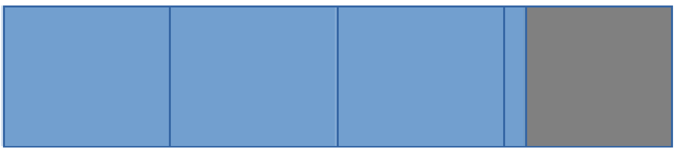


Figure 2: 9 seconds of wasted data [shown in grey]

If conventional buffering is used and playback of the video is stopped after an identical 31 seconds then there is approx 90 seconds of unwatched A/V data in the PC's buffer.

Figure 3: 90 seconds of wasted data [shown in grey - not to scale]

Transmitting data from YouTube's servers to your PC consumes energy, and in the aforementioned example the conventional buffer wasted 81 seconds more A/V data than the minimal buffer of EcoTubeHQ.

81 seconds of A/V data may seem like a tiny amount, but when you consider over 1 billion hours of YouTube videos are watched every day it is clear that the amount of data wasted every day is mind boggling.

A nominal buffer also benefits systems operating under network bandwidth constraints – less data wasted = less bandwidth used.

Automatic Codec Selection

YouTube uses 3 video codecs, av1, vp9 and h.264.

The table below compares the codecs.

Video Codec Comparison

Aspect	AV1	VP9	H.264
Compression Efficiency	Best — ~30% better than VP9, ~50% better than H.264 at same quality	Good — ~25% better than H.264	Baseline — less efficient, requires higher bitrates for equivalent quality
Decoding Speed	Moderate — reasonably fast on modern hardware; hardware decoders increasingly common	Moderate to Good — hardware support widespread	Excellent — hardware support nearly universal; extremely fast

In the **Quality** mode EcoTubeHQ maximizes video quality whilst minimizing data requirements by defaulting to using the codec with the highest compression efficiency along with AMD FSR upscaling:

- Maximum resolution 720p
- AMD FSR upscaling
- av1 is used by default
- If av1 is not available then vp9 is used
- If vp9 is not available then h.264 is used

In the **Powersave** mode EcoTubeHQ focuses on power saving:

- Maximum resolution 480p
- Nearest neighbor upscaling
- vp9 is used by default
- If vp9 is not available then h.264 is used

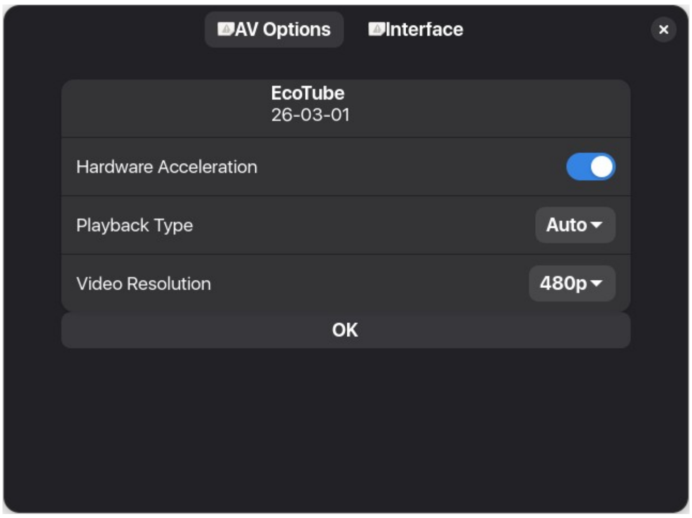
EcotubeHQ further reduces data usage by not outputting 720p video at 60fps as this requires an additional 50%+ of data when compared to 30fps.

Note: YouTube restricts some old h.264 videos to three resolutions: 144p, 360p, and 720p. If EcoTubeHQ is set to output an unavailable h.264 resolution [240p or 480p], it automatically switches to the next highest tier - 240p outputs video at 360p, and 480p outputs video at 720p.

The end result is optimal video quality with minimal data usage.

Player Preferences

Click on the hamburger menu next to the minimize button, then select Preferences. This shows the following…



...here's what it means:

__Hardware Acceleration__

On [Default] - Vulkan GPU Compute Acceleration offloads decoding to your GPU for av1, vp9, h.264, and h.265 video codecs, reducing CPU usage. This improves playback on older PCs and extends laptop battery life. Disable if you experience playback problems.

__Playback Type__

Auto [Default] - **Quality** is the default setting on desktop computers. Laptops use **Quality** when plugged in and automatically switch to **Powersave** on battery power.

Quality - av1 video is upscaled by AMD FSR to optimize video quality. Debanding is also employed to show colors as smooth gradients rather than distinct bands, further improving picture quality.

Powersave - vp9 video uses Nearest Neighbor video upscaling.

__Video Resolution__

Supported Video Resolutions: 720p, **480p [Default]**, 360p, 240p, 144p

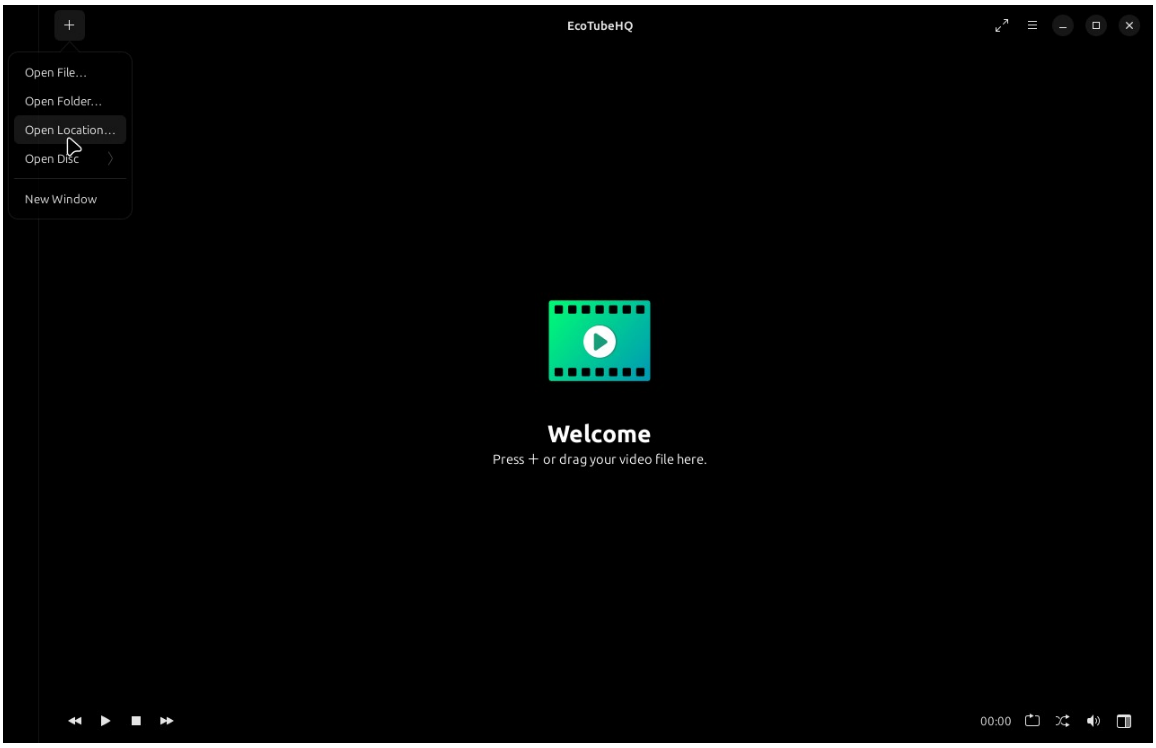
The **Quality** playback mode restricts the maximum video resolution to 720p to minimize data usage, and **Powersave** caps the resolution at 480p to minimize CPU usage.

Note: EcoTubeHQ automatically updates yt-dlp and FFmpeg to minimize issues.

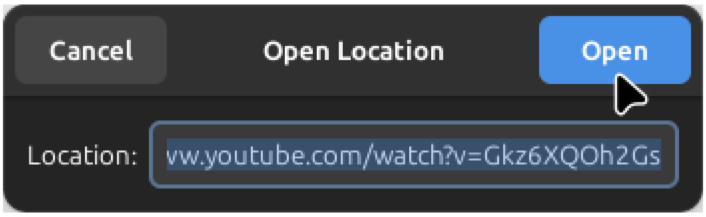
*If the player regularly displays the error '**Playback was terminated abnormally. Reason: loading failed**' when you try and play a video then there may be a problem with the streaming server or the software may need to update. Simply close the player for a while and try again later.*

How to use

To play a video from an online streaming service simply copy the video URL from the browser, click '+' at the top left of the player and select 'Open Location'…



...the player automatically places the copied video location into the 'Location' entry box…



...click the 'Open' button to initiate playback.

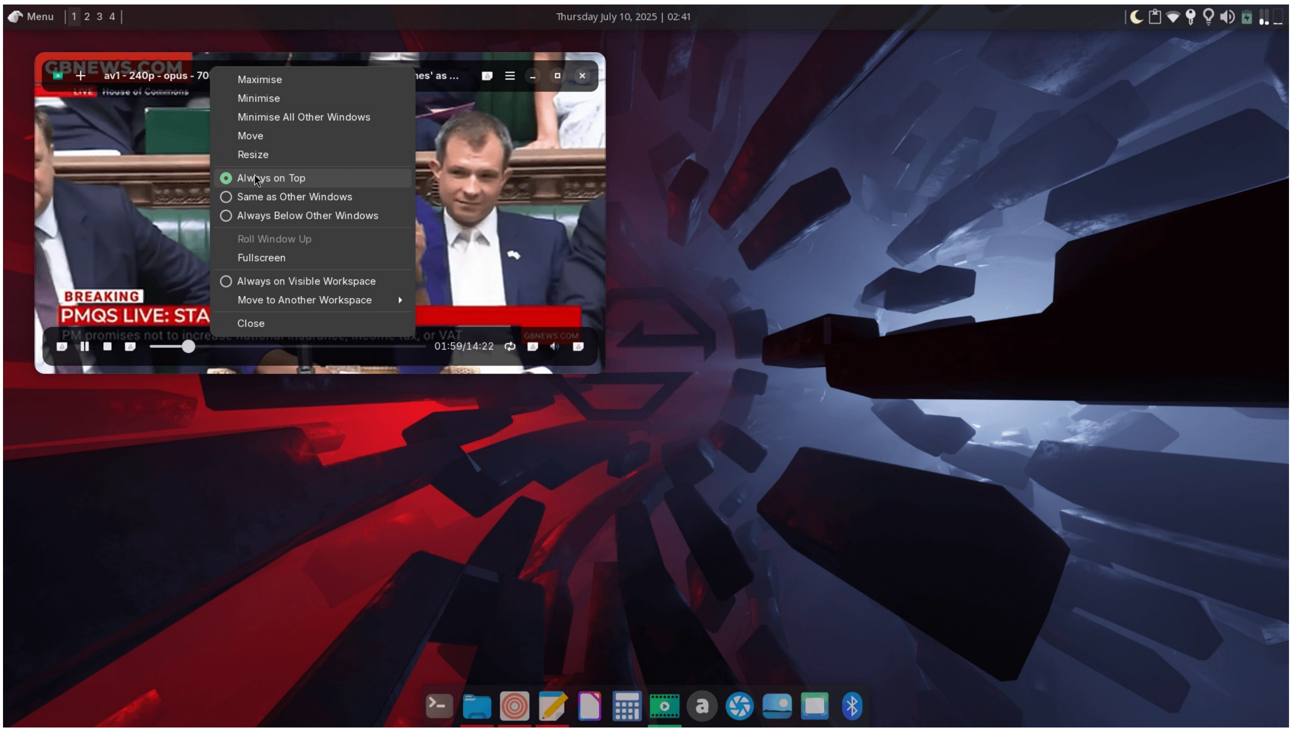
- Press the space bar or right click the mouse on the player screen to pause / play video playback.
- Rotate the mouse wheel on the playback window to turn the audio volume up / down – the volume will reset to normal when the player is restarted.
- Double click the player window during playback to play the video fullscreen.
- Press ‘Esc’ or double click on the player window during fullscreen playback to return the player to non-fullscreen playback.

Miniplayer Usage

If you are watching a YouTube video which does not require fullscreen or you want to minimize CPU usage during playback then set the player's video output resolution to 144p or 240p.

This automatically reduces the size of the player's non-fullscreen window, effectively providing you with a miniplayer. The size of 240p videos is shown in the screenshot below, and 144p videos are displayed in a smaller window.

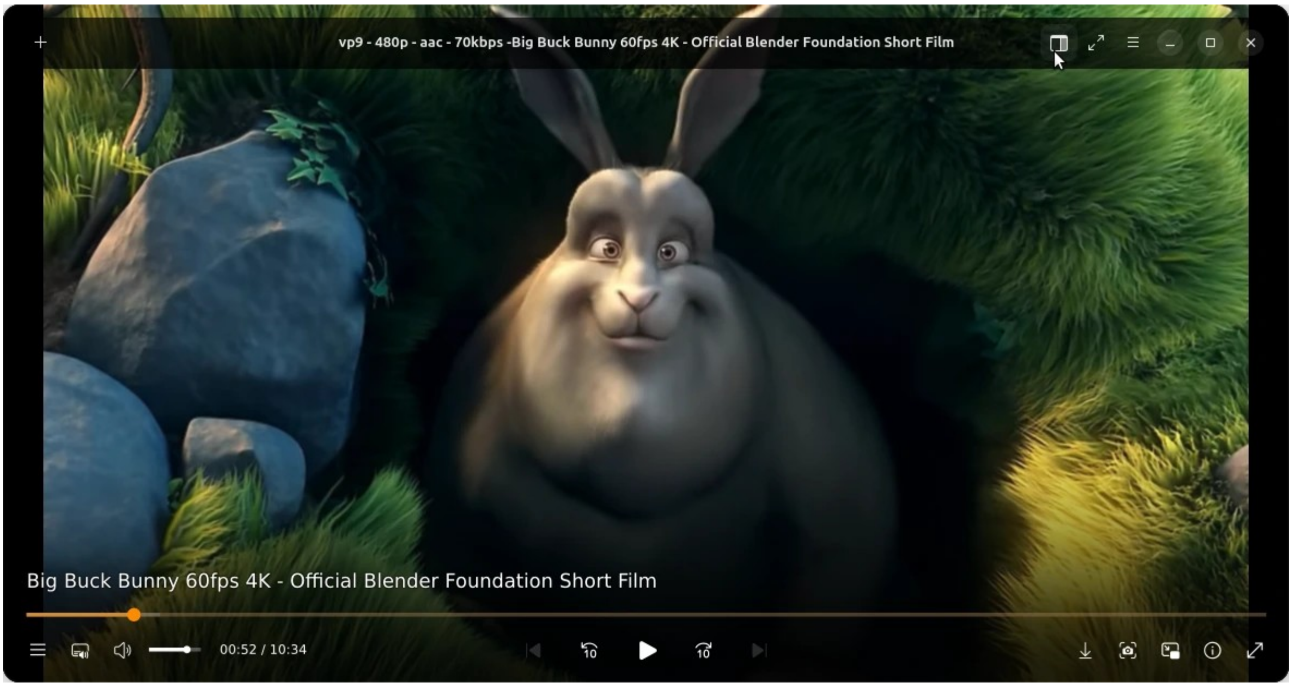
To keep the player on top of all other windows during playback right-click on the top bar and select 'Always on top' as shown below.



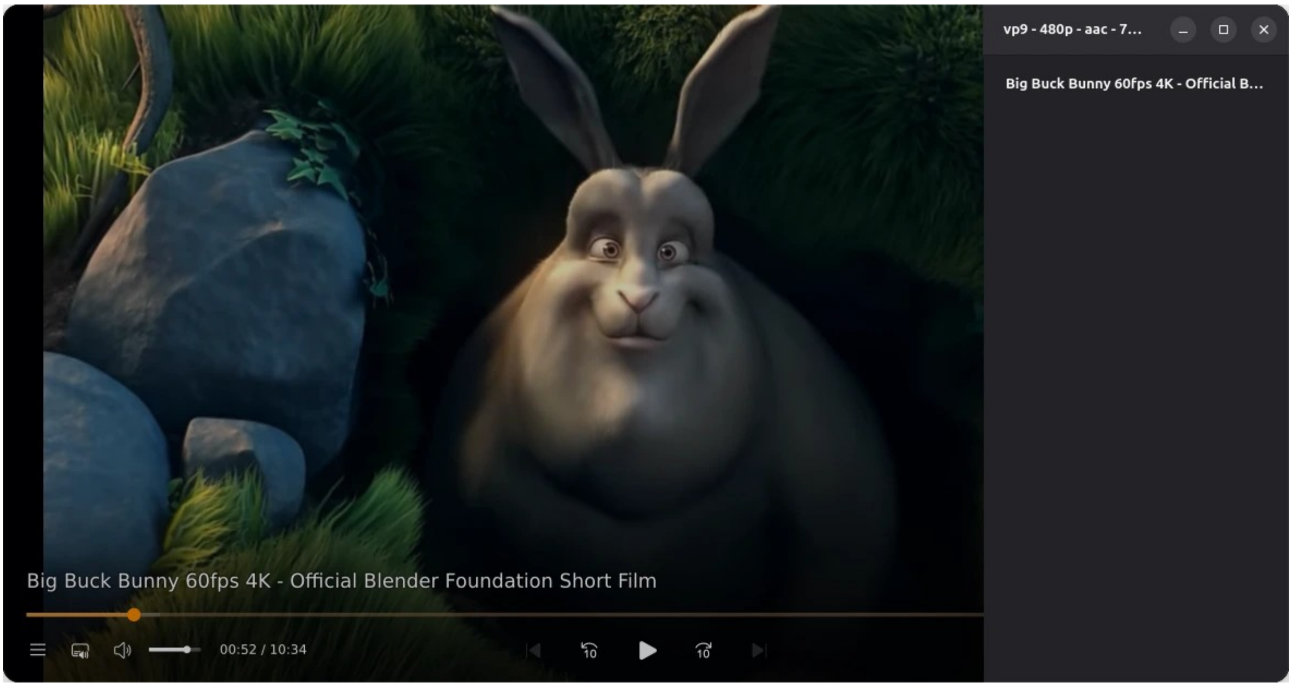
The miniplayer significantly reduces the amount of data used for videos of things such as podcasts where video quality is not paramount.

* To further reduce CPU usage set the Playback Type to **Powersave**

Playlists



To add a video to a playlist click on the top icon indicated above. When this is done the playlist opens.



To add videos from an external streaming service drag and drop the browser URL for the playlist onto the player's playlist. Additional videos can be added prior to, and during playback. To start playback simply double-click on the required playlist video.

Screenshots

To take a screenshot of the on-screen picture click the button at the bottom of the player shown below…

…the image will be stored in the **Pictures** folder.

Audio and video information

To see information relating to the on-screen video click the information button shown below…

EcoTubeHQ Original Concept: Colin Bett

Coding and additional ideas: Sako Adams

Forked from the Celluloid Video Player

Changelog

26.06.02

- Added internal pdf viewer
- Allow 720p in auto mode when laptop is plugged in
- Fix FSR when laptop is plugged in
- Fix resolution fallback when plugged in

26.06.01

- Fix format selection, take into account when audio is not in separate file
- Control end of playback
- Fix playback per block of 10 sec
- Improve UI

26.04.01

- Updated Gnome runtime to the latest version
- Fixed format selection
- Improved video loading

26.03.01

- Updated Preferences and removed HDR option
- Changed video upscale option to nearest neighbor
- Fixed format selection
- Changed app name in Start Menu / launcher

v26.02.01

- Updated preferences and removed deprecated features
- Fixed dark theme issue
- Fixed issue occurring on Wayland on Open Location option
- Auto update ffmpeg

v25.12.02

- Add option to set many seconds of audio/video to prefetch
- Added mpv reload script
- Update ecotube fidelityfx super resolution logo

v25.12.01

- Fixed thumbnail on sandbox
- Disabled video download
- Enhance interface
- Reorganized Auto mode

v25.11.01

- Enhance interface
- Added modern OSC with screenshot and stats
- Fix FSR when plugged in
- Add ‘auto’ mode in desktop instance

v25.10.2

- yt-dlp is auto-updated when a new release is available
- Added modern On Screen Control
- Local files show thumbnails on the seekbar
- Download the video
- Enhance UI

v25.10.1

- Added ‘auto mode’ in playback type
- Fix stream latency
- Enhance UI

v25.9.3

- Updated yt-dlp
- Bilinear upscaling is used for powersave mode
- Enhance UI

v25.9.2

- Fix codec selection in powersave mode
- Updated playlist mode
- Enhance UI

v25.9.1

- Updated playback options - Powersave and Quality
- Updated playlist mode
- Added HDR option in preference
- Enhance UI
- Fixed memory usage in startup

v25.8.2

- Added two playback options - Powersave and Quality
- Removed redundant codec selection
- Reorganized default playlist
- Included no https protocol in format selection
- Reduced network stream usage

v25.8.1

- Forced yt-dlp to use MPEG-DASH on some resolutions
- Reduce stream when paused
- Fixed resolution dropping
- Exclude HDR
- Add playback type

v25.7.2

- Forced mpv to avoid HLS - temporarily

v25.7.1

- Added Vulkan support
- Fixed video / audio download + improved video loading
- Fix format selection
- Updated caching system
- Fixed auto-resize player

v25.5.1

- Upgrade to gtk4.16 and libadwaita-1 1.6
- Enhance interface with playlist overlay
- New update notification
- New dialog for "How to use it"
- Minor playback fix

v25.3.3

- Buffering: prefetch is now limited to 10 seconds to minimize data used when videos are not played in their entirety.
- Hardware acceleration is automatically activated if a laptop is running on battery power.
- Auto update parsers.
- Updated the way default resolutions are selected.
- The preference interface has been updated.

V25.2.1

26 February 2025

Initial conversion from Celluloid.

Thanks to haggen88 for the icon update and albanobattistella for the patches,

14 August 2024
Initial commit to Github.